

C²IM: A Compact Computing-In-Memory Unit of 10 Transistors with Standard 6T SRAM

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Abstract—Memory wall has been a major bottleneck that restrains the speed and power consumption of processors in the Von Neumann architecture. Computing-in-memory (CIM) was proposed as a promising method to tackle the memory wall by implementing computing in memory instead of fetching the value from memory to the processor. Based on the standard 6T-SRAM, this paper proposes a compact CIM (C²IM) unit of 10 transistors. This C²IM unit is capable to not only implement the complete function of SRAM, but also realize the multiplication between the input, which is copied into the unit through the current mirror, and the value stored in the SRAM. Current-mode circuits are adopted in this unit so that it can implement higher energy efficient multiply-accumulate (MAC) operation with simpler control timing and transistor cost. Based on TSMC 65nm CMOS lower power process, the proposed unit can achieve 166.67 TOPS/W energy efficiency under 200MHz clock frequency.

Keywords—computing-in-memory, current-mode, 6T-SRAM

I. INTRODUCTION

Driven by Moore's Law, the performance of the integrated circuits has increased steadily in the past few decades. However, the scaling of memory access time is not consistent with Moore's Law. Therefore, we inevitably hit the memory wall [1], a significant gap between memory and processor in terms of speed and energy consumption [2][3]. Due to the existence of memory wall, the energy cost of the system-on-chip for memory access is a couple of orders-of-magnitude higher than the cost of functional operation. Moreover, most of the popular deep learning algorithms are computing-intensive, and the resulting frequent data movement exacerbates this gap. Computation-in-memory (CIM) is an emerging trend to deal with this bottleneck and has been intensively studied in recent works [3][4][5][6][7]. CIM presents a Non-Von Neumann computation paradigm where computation directly happens inside memory blocks instead of moving out the data before computing. Since multiply-accumulate (MAC) operations and the required memory access dominate the energy consumption in NNs [8], CIM architecture is promising to improve the energy efficiency by reducing data movement.

In conventional CIM architecture, the first step is to convert the inputs to either analog voltage [5] or pulse width modulation (PWM) signals [9]. These analog signals then drive the word line (WL) switches in SRAM array to charge/discharge the bit lines (BLs). Finally, the results on BLs are accumulated or averaged to get the output of MAC operation. By implementing in-memory computing, the total memory access from data host to processor can decrease up to 26.5X [10]. However,

conventional CIM systems often use DACs or time-domain converters (TDCs) as interface to provide input data to NNs. For example, the systems proposed in [6][10][11] use parallel DACs to drive the BLs in SRAM arrays, which results in a significant overhead when the application requires high throughput. Meanwhile, sensors use ADCs to convert original analog information to digital data, which is easier to store in memory. However, this ADC-DAC operation does not contribute to information extraction yet consumes a large portion of hardware resources.

It is well-known that, under the circumstance of limited precision, analog computing has the potential to increase energy efficiency compared with digital counterparts [12][13]. This fact has motivated a range of CIM works to take advantage of in-situ analog computing by exploiting mixed-signal processing architectures. However, the energy consumption of interface (ADC-DAC) and inherent imperfections of analog circuits are two main problems faced by recently proposed CIM works. To tackle these problems, our previous work [14] proposed a near-sensor architecture, applying the binary-weight neural network [15] for classification applications to achieve a better trade-off between model complexity and accuracy. Additionally, we use current-mode circuits to reduce the complexity of MAC operation and gain better scalability compared with previous charge-based designs. In this paper, we present an upgraded unit, namely compact CIM (C²IM), which is more robust.

The remainder of this article is organized as follows. Section II introduces the architecture and operation scheme of the proposed unit in detail. Section III describes the organization of C²IM array. Section IV discusses the simulation results of the proposed design. Finally, the conclusions are presented in Section V.

II. PROPOSED UNIT

The proposed C²IM unit, as shown in Fig. 1, is composed of a standard 6T-SRAM cell and a four-transistors switched-current circuits, which consists of a cascode current mirror and two NMOS switches, ten transistors in total. This C²IM unit is capable to not only implement the complete function of SRAM, but also realize the multiplication between the input and the value stored in the 6T-SRAM. There are two operating stages for the unit, writing stage and computing stage. In the writing stage, the unit can be accessed and written like standard SRAM. And in the computing stage, multiplication is implemented.

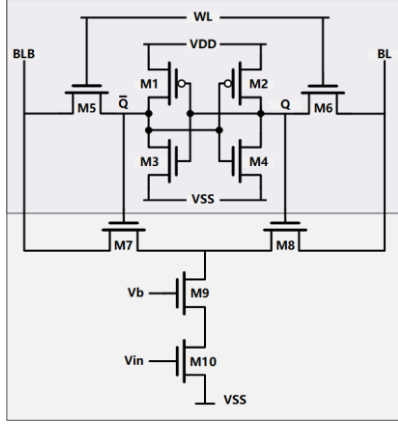


Fig. 1. Overview of the proposed C²IM unit.

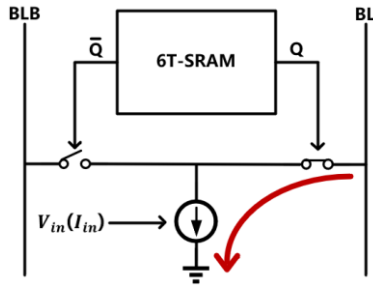


Fig. 2. The simplified model of the C²IM unit in computing stage.

A. Writing stage

In the writing stage, V_b and V_{in} are grounded to avoid the impact of leakage current from the current mirror. In this case, the current mirror is equivalent to an open circuit. Meanwhile, those two switches will not affect the correct read and write operations of SRAM. Therefore, when the C²IM unit is operating in the writing stage, it can be regarded as a standard 6T-SRAM cell, as shown in the upper part of Fig. 1. This six-transistors structure is the most widely used basic circuit in the design of SRAM memory at present, and its read-write principle has been introduced in detail in a large number of documents, so this paper will not repeat it here.

B. Computing stage

When the C²IM unit is operating in the computing stage, the simplified model is shown in Fig. 2. Those two switch transistors are controlled by Q and $\bar{Q}$ of the 6T-SRAM, connecting the current mirror transistors to the BL line or BLB line, respectively. The mirror transistors in the C²IM unit perform as a small current source, which is proportional to the input voltage V_{in} . So, the inputs are fed into the unit in the form of current. In this case, it is more convenient and intuitive to represent the input using current instead of voltage. Therefore, we use I_{in} instead of V_{in} in the following.

According to the of transistor rule in saturation-region, the output current can be given by (1) when taking the channel length modulation into consideration:

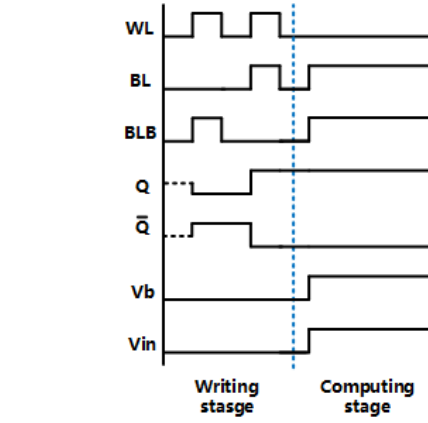


Fig. 3. Operating diagram of the C²IM unit.

$$I_D = \frac{1}{2} \mu_n C_{ox} \frac{W}{L} (V_{GS} - V_{TH})^2 (1 + \lambda V_{BL/BLB}) \quad (1)$$

where the term $\lambda V_{BL/BLB}$ is an input-related gain error caused by channel length modulation effect. To overcome the channel length modulation effect, it is proved that introducing an extra transistor to form the cascode structure works well. Owing to the cascode structure, the channel length modulation can be neglected when the V_b is biased at a proper voltage. Therefore, the current formula of the cascode current mirror is adjusted to equation (2):

$$I_D = \frac{1}{2} \mu_n C_{ox} \frac{W}{L} (V_{GS} - V_{TH})^2 \quad (2)$$

Since Q and $\bar{Q}$ are always in opposite states, there is only one current path in this four-transistors switched-current structure. In fact, the 1-bit weight stored in 6T-SRAM determines the direction of the drain current of the mirror transistors. When the stored value is 1 ($Q = 1$), the transistor controlled by the terminal Q is turned on, and the current flows through the BL line, as shown in Fig. 2. On the contrary, when the stored value is 0 ($Q = 0$), the transistor controlled by the terminal $\bar{Q}$ is turned on, and the current flows through the BLB line.

C. Diagram of operation

Fig. 3 shows the procedure of writing and computing stage. In the writing stage, the SRAM is successively written 0 and 1 in the condition that V_b and V_{in} are grounded. In the computing stage, the word line (WL) is disabled to shield the 6T-SRAM from the disturbing of computing circuits. V_b is biased at a proper voltage to provide a relatively stable drain-source voltage V_{DS} for transistor M10 to weaken the channel length modulation. Meanwhile, V_{in} is connected to the input to force the current mirror to copy the input current into the unit. The bit lines BL/BLB are connected to readout circuits, which pull up the voltage of bit lines. Then, the current path is formed through either the BL line or the BLB line according to the value stored in 6T-SRAM.

In fact, the computing stage of C²IM is to copy an input current, then draw the same amount of current from BL or BLB according to the value stored in 6T-SRAM. This process is essentially a signed weighting of the input current, given by (3):

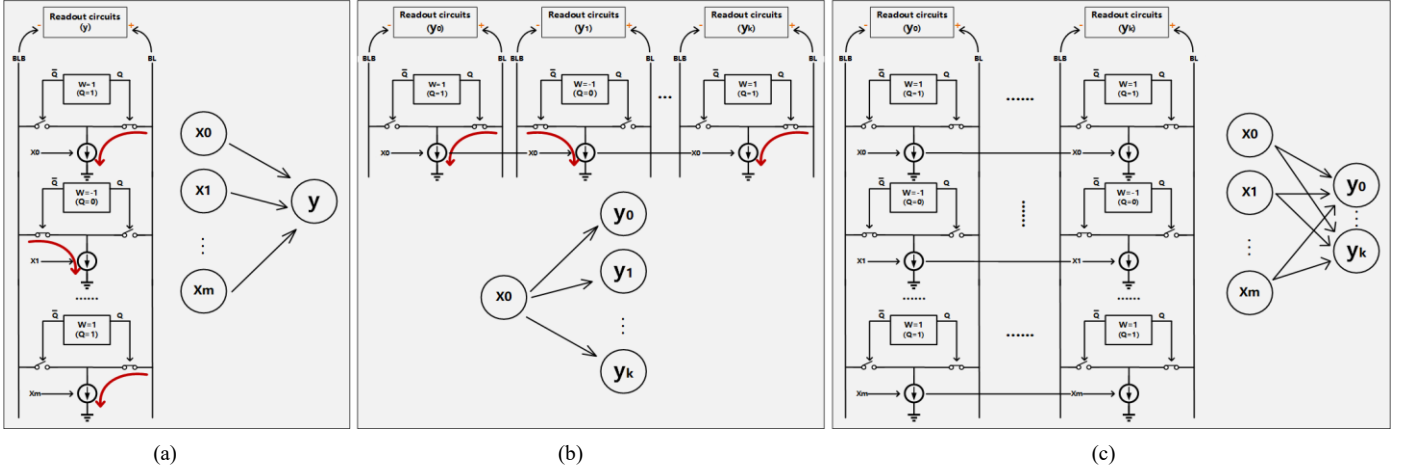


Fig. 4. The topological connection for C^2IM units in a) vertical direction, b) horizontal direction, c) array.

$$y = w \cdot x, w \in \{-1, +1\} \quad (3)$$

where x represents the current signal and w on behalf of weight stored in 6T-SRAM. When the stored value is 1 ($Q = 1$), the weight it represents is +1. Otherwise, the weight it represents is -1 ($Q = 0$).

III. ARRANGEMENT OF ARRAY

According to Kirchhoff's current law (KCL), when the C^2IM units are connected vertically, as shown in Fig. 4 (a), the current formed in computing stage would be summed directly in bit line. In fact, this is one kind of multiply-accumulate (MAC) operation based on current-mode circuits. The advantage of the above circuits is that the speed of computing only depends on the setup time of the current mirror, meaning higher maximum speed.

In the computing stage, only the unit whose current mirror is connected to the BL line or the BLB line through switch transistor contributes current to the corresponding bit line. This process can be modeled by equation (4) and (5).

$$I_{BL} = \sum_{i=0}^m x_i^{w=1} \quad (4)$$

$$I_{BLB} = \sum_{i=0}^m x_i^{w=-1} \quad (5)$$

Then, the subtraction of current between the BL line and the BLB line is implemented in the readout circuits. In other words, this pattern is the multiply-accumulate (MAC) operation of the input vector and stored binary weights, represented by (6).

$$\begin{aligned} y &= I_{BL}^{col} - I_{BLB}^{col} \\ &= \sum_{i=0}^m x_i^{w=1} - \sum_{i=0}^m x_i^{w=-1} \\ &= \sum_{i=0}^m w_i \times x_i \\ &= \vec{w} \cdot \vec{x}, w_i \in \{-1, +1\} \end{aligned} \quad (6)$$

When the C^2IM units are placed in a row, as shown in Fig. 4 (b), the input is shared horizontally. This pattern, k columns represent k channels, corresponds to the single input, multiple outputs in the neural network. It can be seen that when the C^2IM units are organized as an array, the dot product between the input vector and weight matrix would be implemented in a highly parallel manner, as shown in Fig. 4 (c). It is the same way that the fully-connect (FC) layer adopts to implement forward

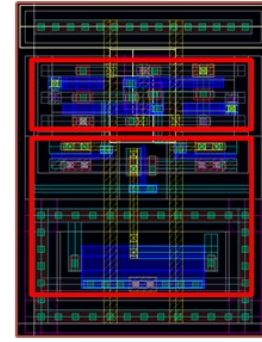


Fig. 5. The layout of the ten-transistors C^2IM unit.

computation. Meanwhile, the forward convolution computation in the convolution layer can also be split into parallel vector-matrix operations, which could be mapped to the C^2IM array.

Note that the MAC operations are implemented by the accumulation of current inside the array and topological connection. The part of storage inside C^2IM unit keeps stationary while in the computing stage, no extra operations needed to be done. Therefore, the data throughput of the MAC operations based on C^2IM array is no longer limited by the traditional SRAM write and read interface speed, but by the number of interfaces m and the number of parallel output channels k of C^2IM array. The number of MAC operations implemented in parallel in each cycle is mk , that is, the throughput of MAC operations of C^2IM array is mk operations/cycle. The operation speed is mkf OPS (operations per second), where f is the frequency of the operation control clock.

IV. SIMULATION AND RESULTS

The proposed C^2IM unit is constructed and simulated based on TSMC 65-nm CMOS lower power process. Fig. 5 shows the layout of the unit, where the upper part is the 6T-SRAM, and the lower part is the 4T switched-current circuit. Obviously, the area of the 4T switched-current circuits is larger than that of 6T-SRAM. The downmost transistor is designed largely to weaken the impact of channel length modulation and keep pretty symmetry for the entire unit. The area of the C^2IM unit is

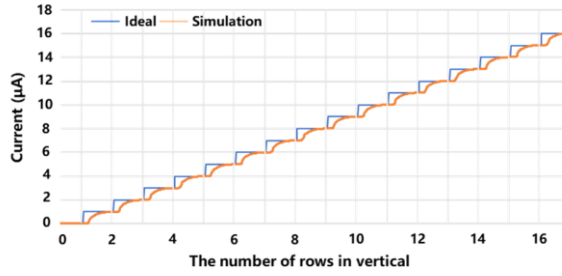


Fig. 6. The results of simulation.

increased by about 60% compared to the standard 6T-SRAM, which is the cost of our introduction of computing circuits.

Table I lists some primary characters of C²IM unit. The unit's power is 1.2 μ W under the circumstance that the input current is 1 μ A. Given the supply voltage and input current, the maximum clock frequency that meets the constraint of the circuit's setup and hold time is 200 MHz. In general, one multiply-accumulate (MAC) operation is done at one cycle inside the unit. Therefore, at the maximum operating frequency of 200MHz, the throughput of the unit is 200 MOPS. The energy efficiency that the unit can achieve is 166.67 TOPS/W.

The linearity shown by the proposed C²IM unit in the computing stage is very important to the accuracy of the multiply-accumulate (MAC) operation. In order to verify the linearity, we could observe the change of the output current in the bit line when the number of units connected in one column varies. The connection pattern in Fig. 4 (a) caters to the verification requirements. In order to see the current change more intuitively, we add one unit at each time to the column, so that the current output should have a linear relationship with the number of units. The simulation is carried out under 200 MHz clock frequency and 1 μ A input current for all units. The results of the simulation are shown in Fig. 6. The current mirror needs a setup time to provide the copied input current, which explains why the simulation waveform lags behind the ideal waveform. Before the end of each cycle, the simulation waveform will catch up with the ideal waveform. The simulation results show that the linearity of the proposed C²IM unit is competent to multiply-accumulate (MAC) operation.

TABLE I. BASIC CHARACTERS

Technology	TSMC 65-nm 1P9M
Supply voltage	1.2 V
Input current	1 μ A
Area	12.82 μ m ²
Max. frequency	200 MHz
Power	1.2 μ W
Throughput	200 MOPS
Energy efficiency	166.67 TOPS/W

V. CONCLUSION

This paper proposes a compact and simple architecture of computing-in-memory, namely C²IM unit. This C²IM unit is

capable to not only implement the complete function of SRAM, but also realize the multiplication between the input, which is copied into the unit through the current mirror, and the value stored in the SRAM. Based on the current-mode computing circuit, accumulation is easy to implement according to the topological connection. Up to 166.67 TOPS/W energy efficiency the proposed C²IM unit can achieve under 200 MHz clock frequency.

ACKNOWLEDGMENT

The authors would like to acknowledge supports from National Natural Science Foundation of China under grant No. 91648116. The authors would also acknowledge support from Beijing Innovation Center for Future Chips, Tsinghua University.

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